



Database Forensics

How Data Solves Crimes in Large Organisations

**A 6-slide guide to investigating data
at scale with SQL and audit logs**

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DATABASE FORENSICS = DIGITAL EVIDENCE

FORENSIC RECONSTRUCTION OF DATABASE

PART ONE

A company found a \$12,000 overpayment. The payroll shows no anomaly. HR said, “ Nothing was deleted.”

Rebuilding the truth from deleted data. (How we recover what happened, when, and by who - even after deletion)

DELETION

1. Logical deletion (deleted = yes) hidden
2. Physical deletion(logged) harder, riskier.

CORE (DATA LIFECYCLE IN SQL)

1. SELECT: How you see data. Baseline for normal
2. UPDATE: How fraud hides data. Data stays, meaning changes.
3. DELETE: How data gets removed. Gone from table, but logged
4. DROP: How the data gets erased. Nuclear option, heavy footprint.

QUESTION: If the record is gone from the UI where do we find it?



1. **SELECT**: This is just reading data. Nothing Changes

```
SELECT * FROM payroll
```

IF the data is there **you see it**. if it is gone **you don't**.

UPDATE	DELETE	DROP
Logical deletion	Physical deletion	Physical deletion
UPDATE payroll set status = 'deleted' where emp_id = "102";	DELETE FROM payroll WHERE emp_id = 102;	DROP TABLE payroll;
If you can still see the row with WHERE status = 'deleted' it was UPDATE . Logical	If the row is gone but you see it in audit logs it was DELETED . Physical	If the whole Table is gone, it was DROP . Nuclear option

SELECT shows you what'S there. UPDATE hides it in plain sight. DELETE removes it. DROP erases the whole thing. As a forensic investigator your job is to know where to look after each one.



COMMAND	WHAT IT DOES	IS IT LOGICAL OR PHYSICAL	CAN WE QUERY IT BACK?	HOW YOU RETRIEVE IT
UPDATE	Modifies existing rows. Data stays in place.	Logical	Yes, directly	SELECT * FROM table WHERE status='deleted'
DELETE	Removes the row from the table. Table structure stays.	Physical	No, not directly	Check audit logs and restore from backup
DROP	Deletes the entire table and structure. Nothing stays.	Physical	No	Only from backup. Always logged (craving the unallocated disk space)

SIMILARITIES

1. All 3 are DML/DDL commands that change data
2. All 3 get logged in audit trails if logging is on
3. All 3 are used in fraud to hide evidence.

TAKEAWAY: Fraudsters love **UPDATE** because it's quiet. **DELETE** and **DROP** are loud and leave traces.

As a forensic investigator, **UPDATE** is your easiest win, **DELETE** is your log-diving job, and **DROP** is your "someone panicked" signal.

DDL

1. CREATE
2. ALTER
3. DROP
4. TRUNCATE

DML

1. INSERT
2. UPDATE
3. DELETE

WHERE 90% OF FRAUD OCCURS



CRIME SCENE

```
-- Creat the table
CREATE TABLE tbl_Payroll (
  Emp_ID INT,
  Employee_Name VARCHAR(50),
  Salary_USD INT
);
-- Insert the target data
INSERT INTO tbl_Payroll VALUES (101, 'Alice' , 8500) ;
INSERT INTO tbl_Payroll VALUES (102, 'bob' , 6200) ;
INSERT INTO tbl_Payroll VALUES (103, 'Charlie' , 9100) ;
```

ACTIVE TABLE: tbl_Payroll

`SELECT` * tbl_Payroll

Emp_ID	Employee_Name	Salary_USD
101	Alice	8500
102	bob	6200
103	Charlie	9100

BEFORE

Emp_ID	Employee_Name	Salary_USD
101	Alice	8500
102	bob	6200
103	Charlie	9100

COMMAND

```
EXECUTE: DELETE FROM
tbl_Payroll;
WHERE Emp_ID=103;
```

AFTER

Emp_ID	Employee_Name	Salary_USD
101	Alice	8500
102	bob	6200

BEFORE

Emp_ID	Employee_Name	Salary_USD
101	Alice	8500
102	bob	6200
103	Charlie	9100

COMMAND

```
EXECUTE: UPDATE  
tbl_Payroll;  
SET Status = 'Deleted'  
WHERE Emp_ID=103;
```

AFTER

Emp_ID	Employee_Name	Salary_USD
101	Alice	8500
102	bob	6200

BEFORE

Emp_ID	Employee_Name	Salary_USD
101	Alice	8500
102	bob	6200
103	Charlie	9100

COMMAND

```
EXECUTE: DROP TABLE  
tbl_Payroll;
```

AFTER

ERROR: Table 'tbl_Payroll' does not exist

UPDATE

```
SELECT * FROM tbl_Payroll WHERE Status = 'deleted';
```

```
SELECT Emp_ID , Employee_Name, Salary_USD  
FROM tbl_Payroll  
WHERE Status = 'Active';
```

```
SELECT * FROM tbl_Payroll;
```

Emp_ID	Employee_Name	Salary_USD	STATUS
101	Alice	8500	Active
102	bob	6200	Active
103	Charlie	9100	Deleted